

1. Show that NP is closed under union and concatenation.

We have two following languages L_1 and L_2 , and TM M_1 and M_2 are the Non-Deterministic Turing Machines that decides them in poly time for both Union and Concatenation.

1. Union, a non-deterministic Turing machine L that decides the union of L_1 and L_2 .

TM L = "On input x

- Non-Deterministically decide whether to check if x is a member of L_1 or L_2 .
- If L_1 was decided, run M_1 on x , and output M_1 's decision.
- If L_2 was decided, run M_2 on x , and output M_2 's decision."

TM L will run in $O(1) + \text{MAX}(O(M_1), O(M_2)) = \text{MAX}(O(M_1), O(M_2))$, which must be in polynomial time, since M_1 and M_2 run in polynomial time. Because TM L decides $L_1 \cup L_2$ in polynomial time, $L_1 \cup L_2$ is in NP.

2. In concatenation, a non-deterministic Turing machine L that decides the concatenation of L_1 and L_2 . In this construct, languages L_1 and L_2 contain strings that start with a string that is in L_1 and end with a string that is in L_2 .

TM L = "On input x

- Non-Deterministically divide x into y and z .
- Run M_1 on y .
- Run M_2 on z .
- If both M_1 and M_2 accepted, accept; else reject."

TM L will run in $O(1) + O(M_1) + O(M_2) = \text{MAX}(O(M_1), O(M_2))$, which must be in polynomial time, since M_1 and M_2 run in polynomial time. Because TM L decides $(L_1 \text{ concatenated with } L_2)$ in poly time $(L_1 \text{ concatenated with } L_2)$ is in NP.

2. Let $\text{DOUBLE-SAT} = \{f \mid f \text{ has at least two satisfying assignments}\}$. Show that DOUBLE-SAT is NP-complete. I would recommend reducing, naturally, from SAT.

$\text{DOUBLE-SAT} \in \text{NP}$ because simply guesses two different assignments to all variables and verify that each clause is satisfied in both cases. The following nondeterministic polynomial time machine can guess two assignments and accept if both assignments satisfy f on input f .

Now, to prove that DOUBLE-SAT is NP-complete I will try to reduce it to SAT by showing that $\text{SAT} \leq_p \text{DOUBLE-SAT}$. The following TM M computes the polynomial time reduction M .

$M = \text{"On input } \langle f \rangle, \text{ a Boolean formula with variables } x_1, x_2, \dots, x_n:$

- Let f' be $f \wedge (x \text{ union } x')$, where x is a new variable.
- Output $\langle f' \rangle$."

If $f \in \text{SAT}$, then f' has at least two satisfying assignments that can be obtained from the original satisfying assignments of f by changing the value of x . If $f' \in \text{DOUBLE-SAT}$, then f is also satisfiable because x does not appear in f . Therefore, $f \in \text{SAT}$ if and only if $f(f) \in \text{DOUBLE-SAT}$.